

- Filename: comptia-pentest-plus-pt0-002-3-3-1-arp-and-dns-poisoning.md
- Show Name: Pentest+ (PT0-002)
- Topic Name: Attacks and Exploits
- Episode Name: ARP and DNS Poisoning

ARP and DNS Poisoning

Objectives:

- Define, explain, and demonstrate ARP and DNS poisoning attacks
-

- How does ARP work?
 - Resolves IP to MAC
 - Helps hosts find other hosts
 - ARP Broadcasts
 - Wireshark demo (Who has this IP? Tell this device)
 - ARP Table
 - Dynamic ARP
 - `arp -a`
- Attacking ARP
 - ARP Cache Poisoning or ARP Spoofing Attack
 - Attacker can go after the switch
 - Attacker can go after the host
 - Threats
 - Sniffing
 - MitM
 - Session Hijacking
 - DoS
- ARP Poisoning Tools
 - BetterCAP
 - Ettercap - DEMO
 - Run Wireshark from terminal first
 - Then show how you can't see telnet login
 - Now poison ARP with ettercap
 - Then show how we can now view telnet login
 - dsniff
 - arpspoof
- DNS basics
 - Resolve domains to IP
 - Windows DNS Lookup order
 - Checks self (am I the device I'm looking for?)
 - Checks DNS Resolver Cache

- Checks the Hosts file (do I already know where this is?)
 - Checks with DNS Servers
- DNS Attacks
 - Modifying hosts DNS info
 - Tricks hosts to query malicious DNS
 - Host file entries
 - Malicious proxy
- DNS cache poisoning
 - Tricking clients into thinking that attacker is legit DNS
 - Redirect targets to malicious sites
 - **DEMO: Ettercap for DNS poisoning**
 1. Modify `etter.dns` file to have fake A records
 - Copy from Kali `/Tools/fakeArec.txt`
 - Change IP to match IP of bWAPP
 2. Run ettercap GUI
 - Scan for hosts
 - Add DNS server IP(192.168.241.1) to Target1
 - Add victim IP(192.168.241.x) to Target2
 - Ettercap Menu(3 dots) > Plugins > Manage Plugins > dns_spoof
 - MITM menu > ARP Poisoning
 3. ping `www.facebook.com` from target
 - `ipconfig /displaydns`